

October 23, 2025

Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics Engineering

Lesson title:

Bionic locomotion principles – Inchworm and crawling locomotion

Prof. Donato Romano (donato.romano@santannapisa.it)



Larval legged locomotion

Endopterygote insects show complete metamorphosis, including a distinct larval stage.

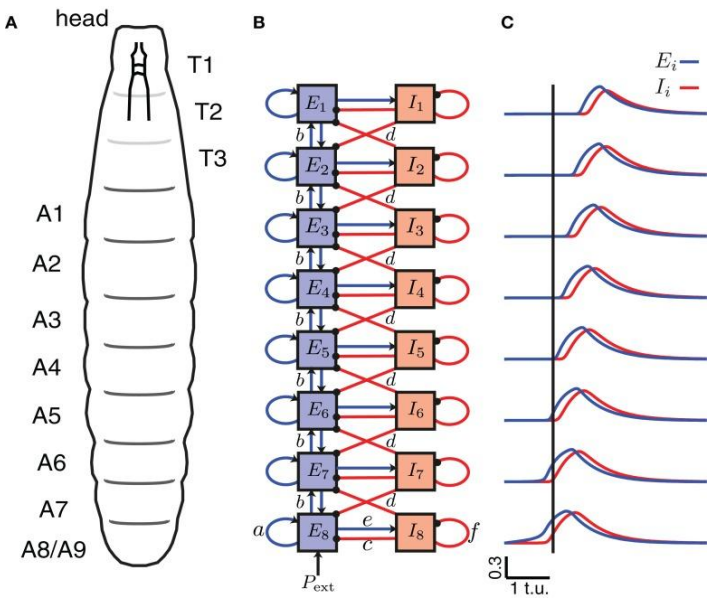
There are four main types of endopterygote larvae types:

1. Apodous larvae: no legs at all and are poorly sclerotized. Based on sclerotization, three apodous forms are recognized:

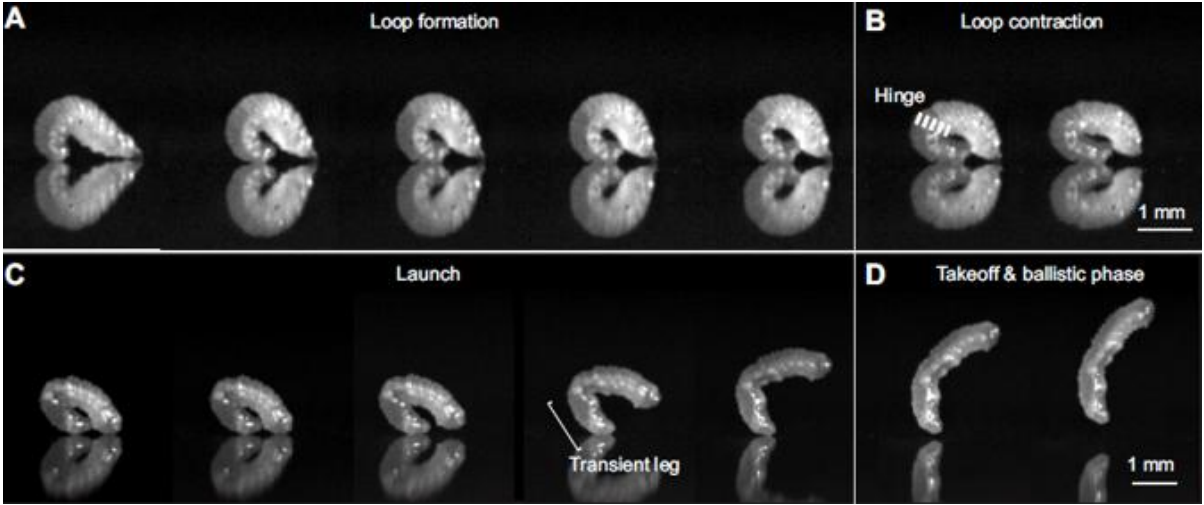
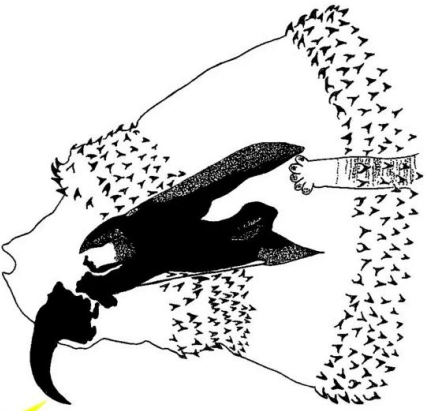
- Eucephalous – with well sclerotized head capsule. Found in Nematocera, Buprestidae and Cerambycidae families.
- Hemicephalus – with a reduced head capsule, retractable into the thorax. Found in Tipulidae and Brachycera families.
- Acephalus – without head capsule. Found in Cyclorrhapha.



Acephalus larvae planar locomotion and jumping



A network model for peristaltic wave propagation with eight interconnected segments, each consisting of an excitatory (E) and an inhibitory (I) neuronal population. Connections a, c, e, and f are shown only for segment A8, but are found in each segment. The segments are connected with nearest-neighbor excitatory b (blue, arrows), and inhibitory d (red, dots), connections. Forward waves are initiated by providing external input P_{ext} into the excitatory population of segment A8. (C) The excitatory activity for each of the eight segments is shown in blue, and the inhibitory activity in red during a forward wave as a function of time.



Larval legged locomotion

Endopterygote insects show complete metamorphosis, including a distinct larval stage.

There are four main types of endopterygote larvae types:

2. Protopod larvae: larvae have many different forms and often unlike a normal insect form. They hatch from eggs which contain very little yolk. E.g. first instar larvae of parasitic hymenoptera.

3. Polypod larvae: also known as eruciform larvae, these larvae have abdominal **prolegs**, in addition to usual thoracic legs. They are poorly sclerotized and relatively inactive. The best example is caterpillars of lepidopterans.

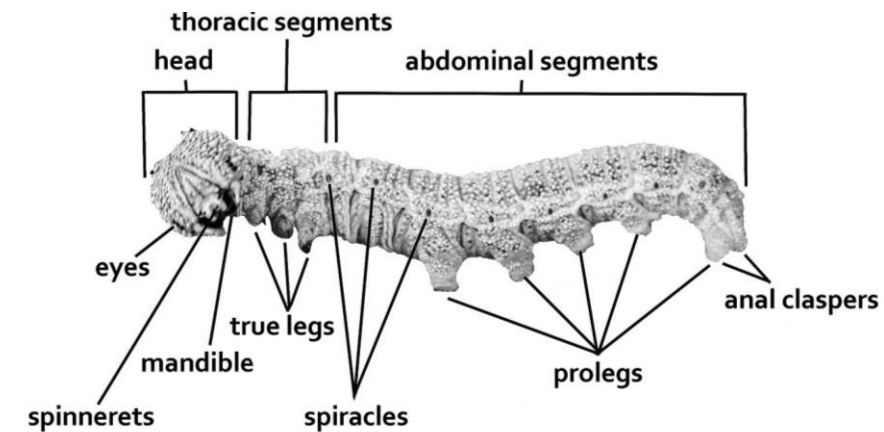
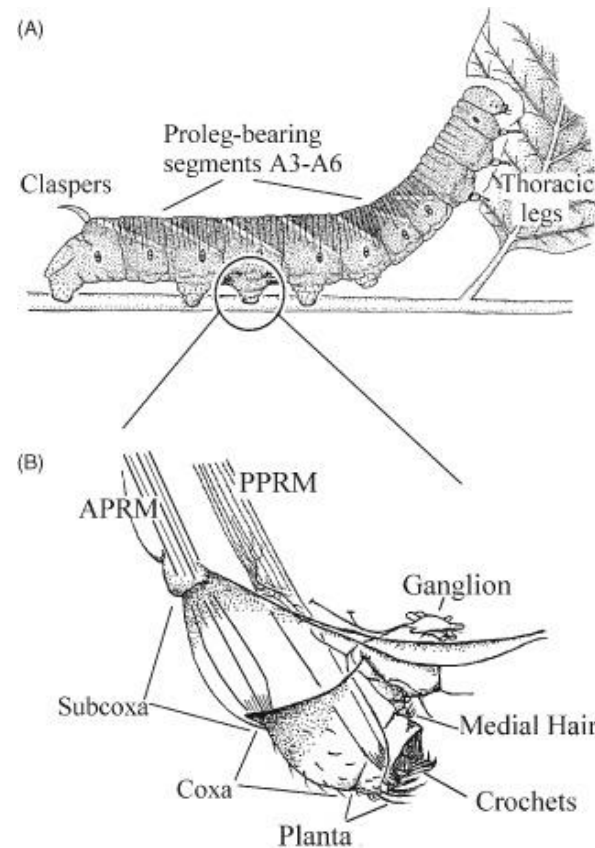


Larval legged locomotion

Prolegs

A proleg is a small, fleshy, stub structure found on the ventral surface of the abdomen of most larval forms of insects of the order Lepidoptera, though they can also be found on other larval insects such as sawflies and a few other types of insects. In all the orders in which they appear, mainly Hymenoptera and Lepidoptera, prolegs of any form evolved independently of each other by convergent evolution.

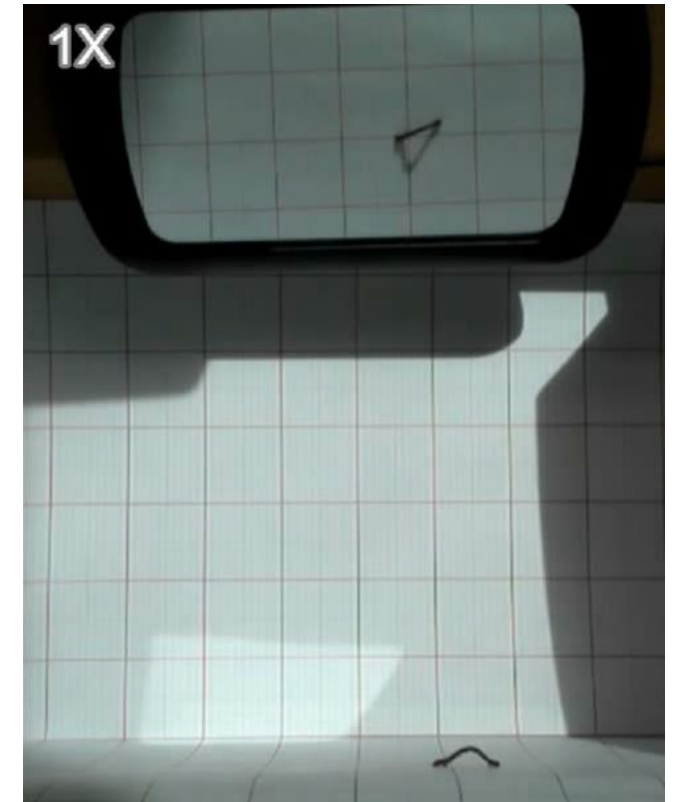
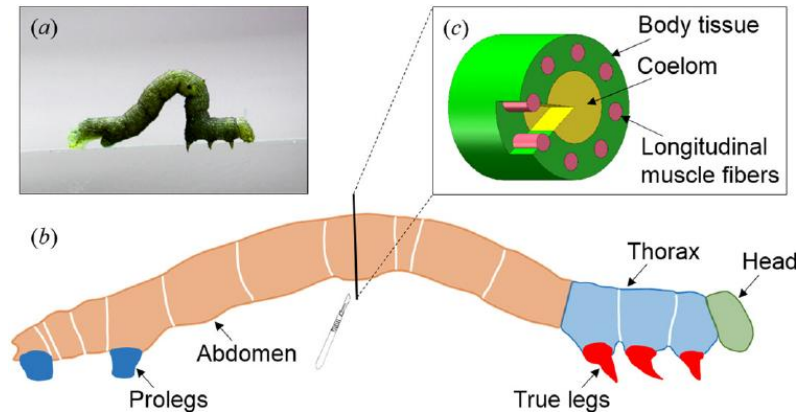
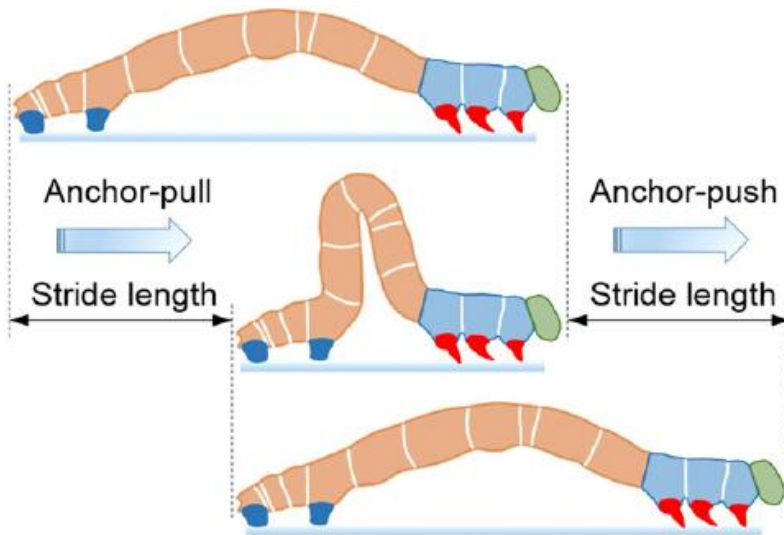
Prolegs of lepidopteran larvae have a small circle of gripping hooks, called "**crochets**". The arrangement of the crochets can be helpful in identification to family level. Although the point has been debated, prolegs are not widely regarded as true legs, derived from the primitive uniramous limbs. Certainly in their morphology they are not jointed, and so lack the five segments (coxa, trochanter, femur, tibia, tarsus) of thoracic insect legs. Prolegs do have limited musculature, but much of their movement is hydraulically powered.



Larval legged locomotion

Prolegs

Caterpillars called "loopers", "spanworms", or "inchworms" due to their characteristic looping gait resulting from the lack of the full complement of prolegs. Since they have no legs in the middle portion of their body, they need to pull their hind portion forward forming a loop, and then move forward with the front portion of their body.

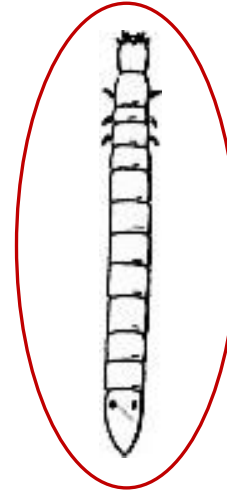
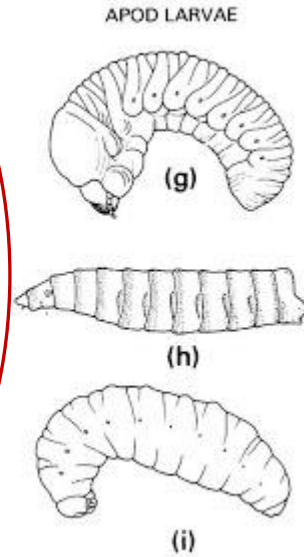
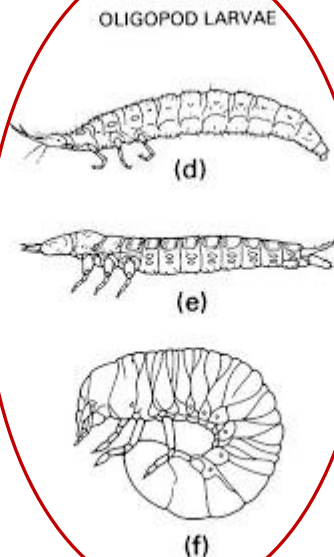
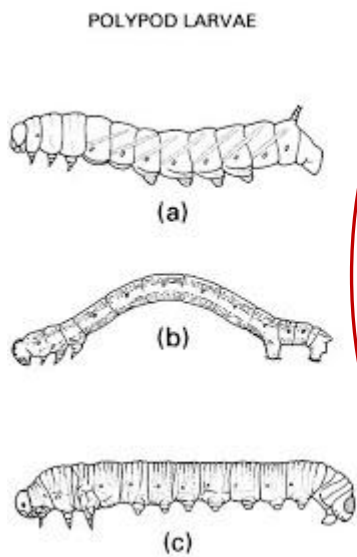


Larval legged locomotion

Endopterygote insects show complete metamorphosis, including a distinct larval stage.

There are four main types of endopterygote larvae types:

4. Oligopod larvae: have well developed head capsule and mouthparts are similar to the adult, but without compound eyes. They have six legs. No abdominal prolegs. Several types can be seen: **Campodeiform** – well sclerotized, dorso-ventrally flattened body. Usually long-legged predators with prognathous mouthparts. **Scarabeiform** – poorly sclerotized, flat thorax and abdomen. Usually short legged and inactive burrowing forms. **Elateriform**. **Oniscriform**.

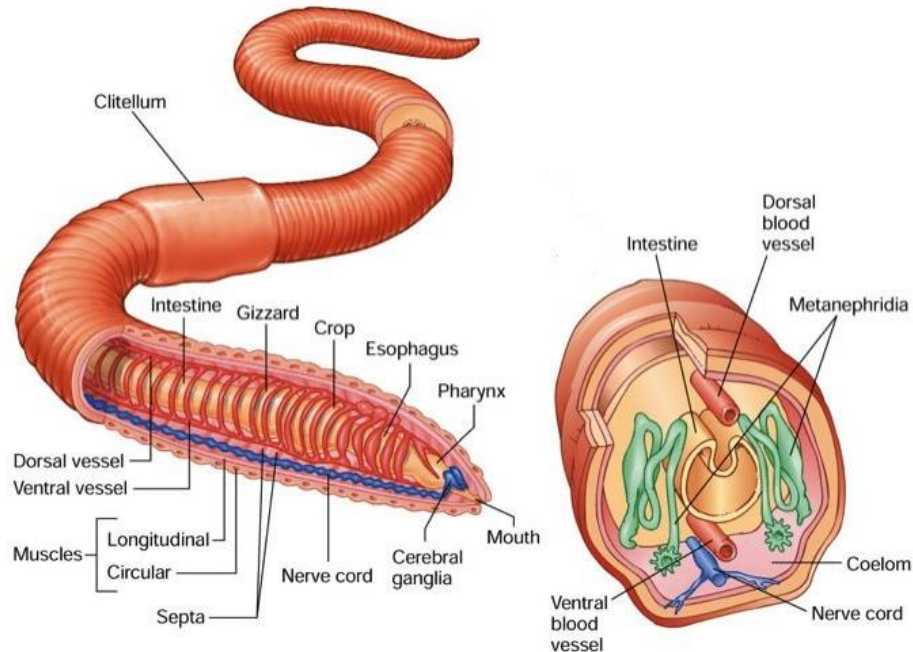
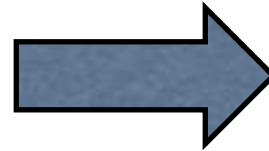


Locomotion in earthworms

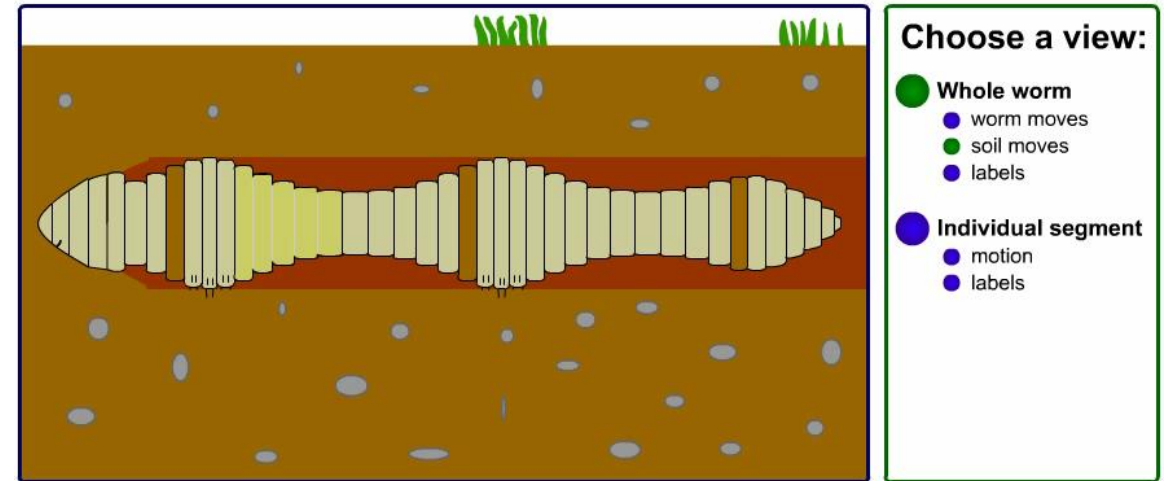
Oligochaeta

Phylum: Annelida

Class: Clitellata



Form and locomotion of an earthworm (class Oligochaeta, phylum Annelida)



Earthworms spend most of their time below ground ingesting soil and decaying plant material, and contribute significantly to the health of terrestrial ecosystems. To move, they squeeze the anterior end forward into the soil by pushing against other parts of the body that serve as anchors. The anterior segments then expand as a new anchor that helps pull forward segments behind them. A simple system of circular and longitudinal muscles, and a hydrostatic skeleton, create these wave-like movements.

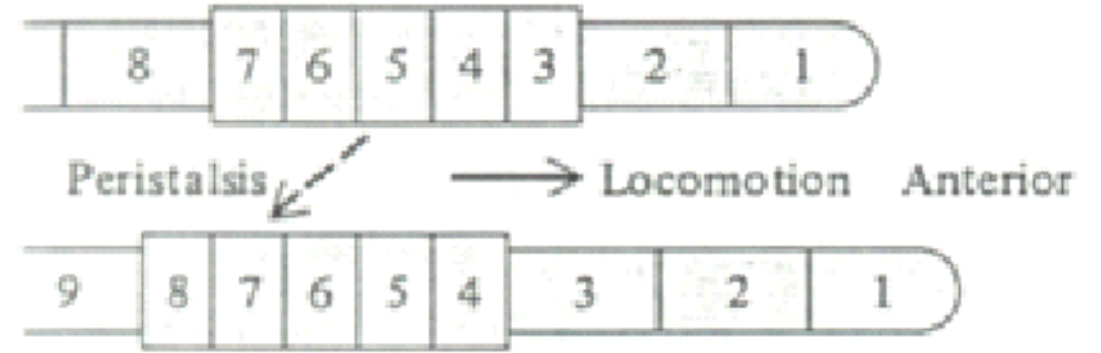
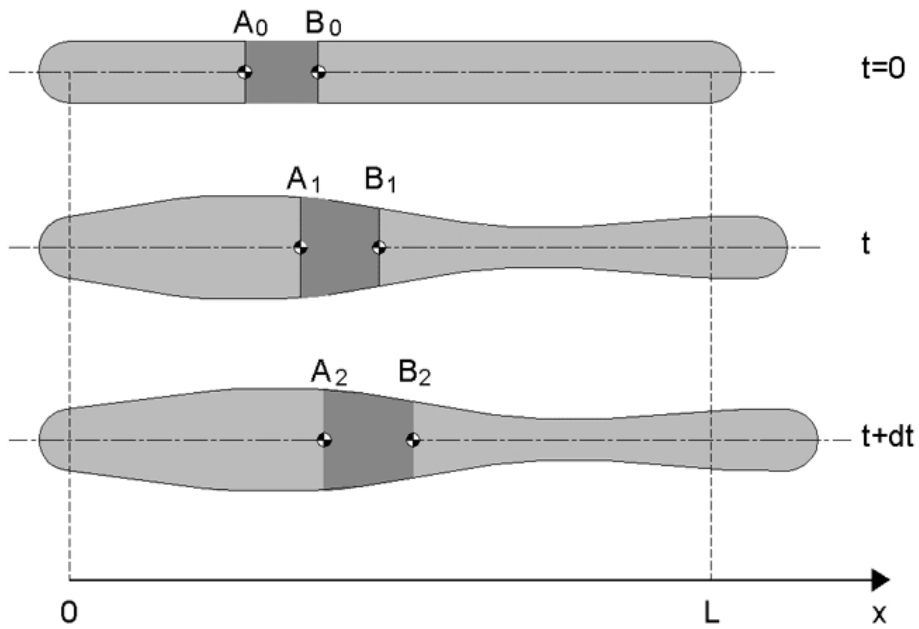
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Heather Kroening
August 2000

Hydrostatic skeleton with longitudinal and circular muscles, producing undulatory motion by peristaltic changes in body shape

From the animal to the mathematics



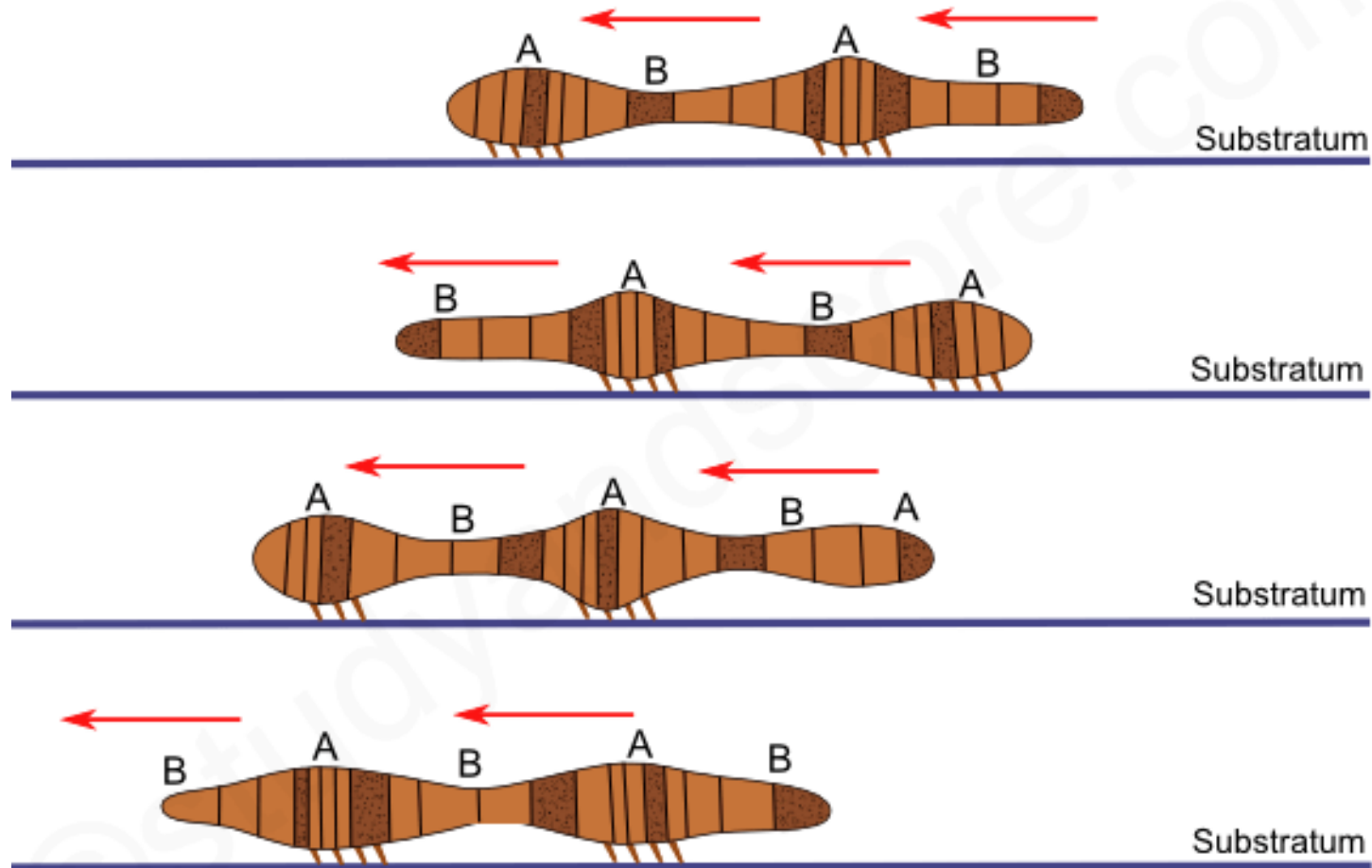
Earthworm locomotion exploits two kinds of waves along the slender body:

- longitudinal (axial strain)
- transversal (radial strain)

Locomotion in earthworms

A- Contraction of longitudinal muscles and relaxation of circular muscles (Setae protruded)

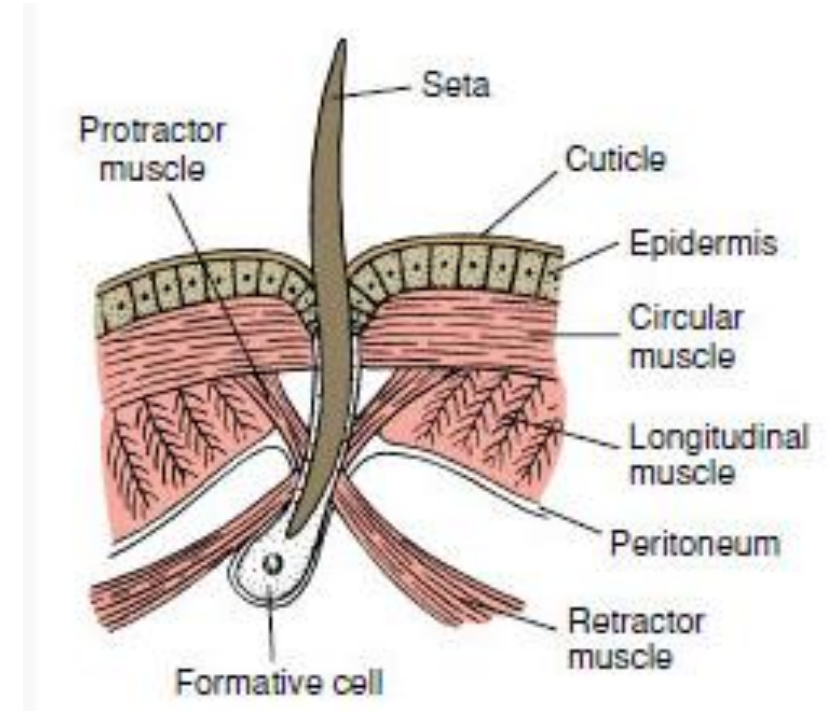
B- Contraction of circular muscles and relaxation of longitudinal muscles (Setae withdrawn)



Locomotion in earthworms

Each segment has four pairs of chitinous setae (in some species each segment may bears up to 100 or more setae). The seta is a bristlelike rod set in a sac within the body wall that is moved by tiny muscles.

Setae are used to anchor parts of the body to prevent slipping during locomotion.



Burrowing locomotion - Physical properties of soils

Texture: proportion of the three kinds of soil mineral particles (sand, silt, and clay)

Structure: when iron oxides, carbonates, clay, silica and humus, coat particles adhere into larger, and relatively stable secondary structures.

Porosity: void part of the soil volume that is occupied by gases/water.

Consistency: the ability of soil materials to stick together.

Resistivity: the resistance to conduction of electric currents (in general with values from 10 up to 1000 Ω -m).

Apparent density: a property of "divided" solids (powders, granules, etc.).

It is the mass of the many particles of the material divided by the total volume they occupy (including particle volume, inter-particle void volume, and internal pore volume). While **particle density (real density)** does not include the volume of voids between particles. Bulk density is not an intrinsic property of a material, but it can change depending on how the material is handled.

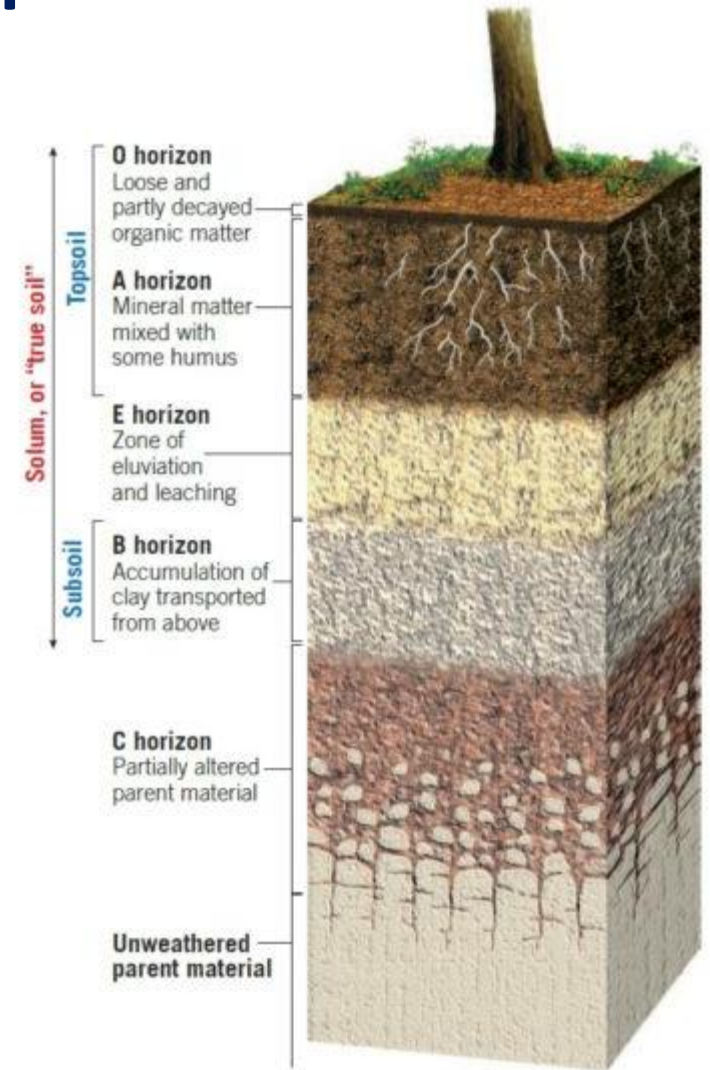
Dry apparent density

$$\rho_b = \frac{M_s}{V_t}$$

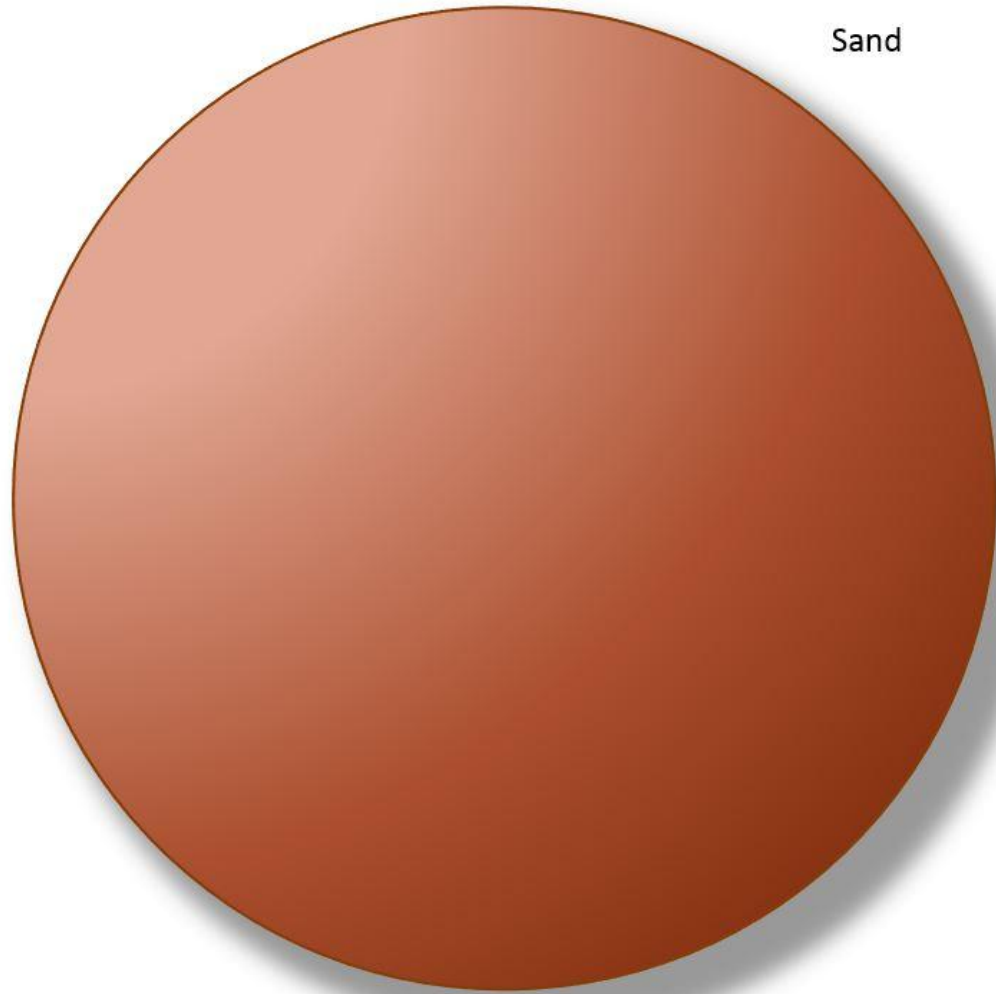
Wet (total) apparent density

$$\rho_t = \frac{M_t}{V_t}$$

Where $M_t = M_s + M_l$

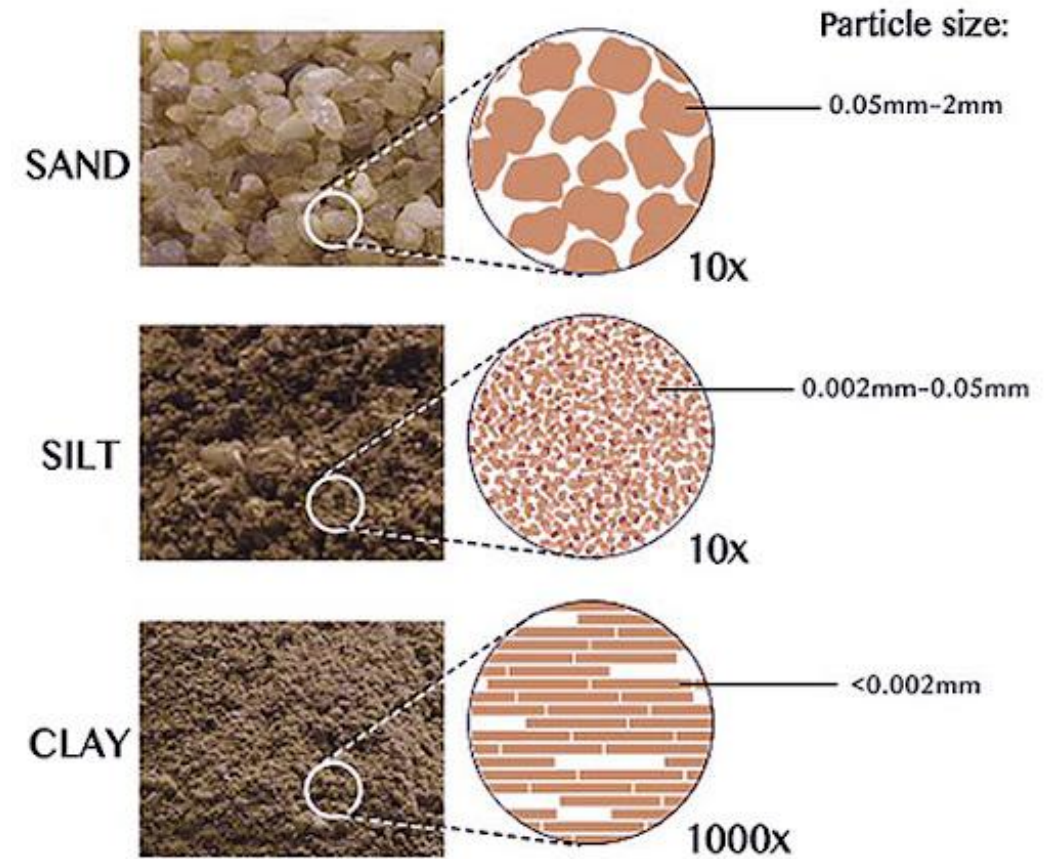


Physical properties of soils



Sand

- Silt
- Clay (× 10)

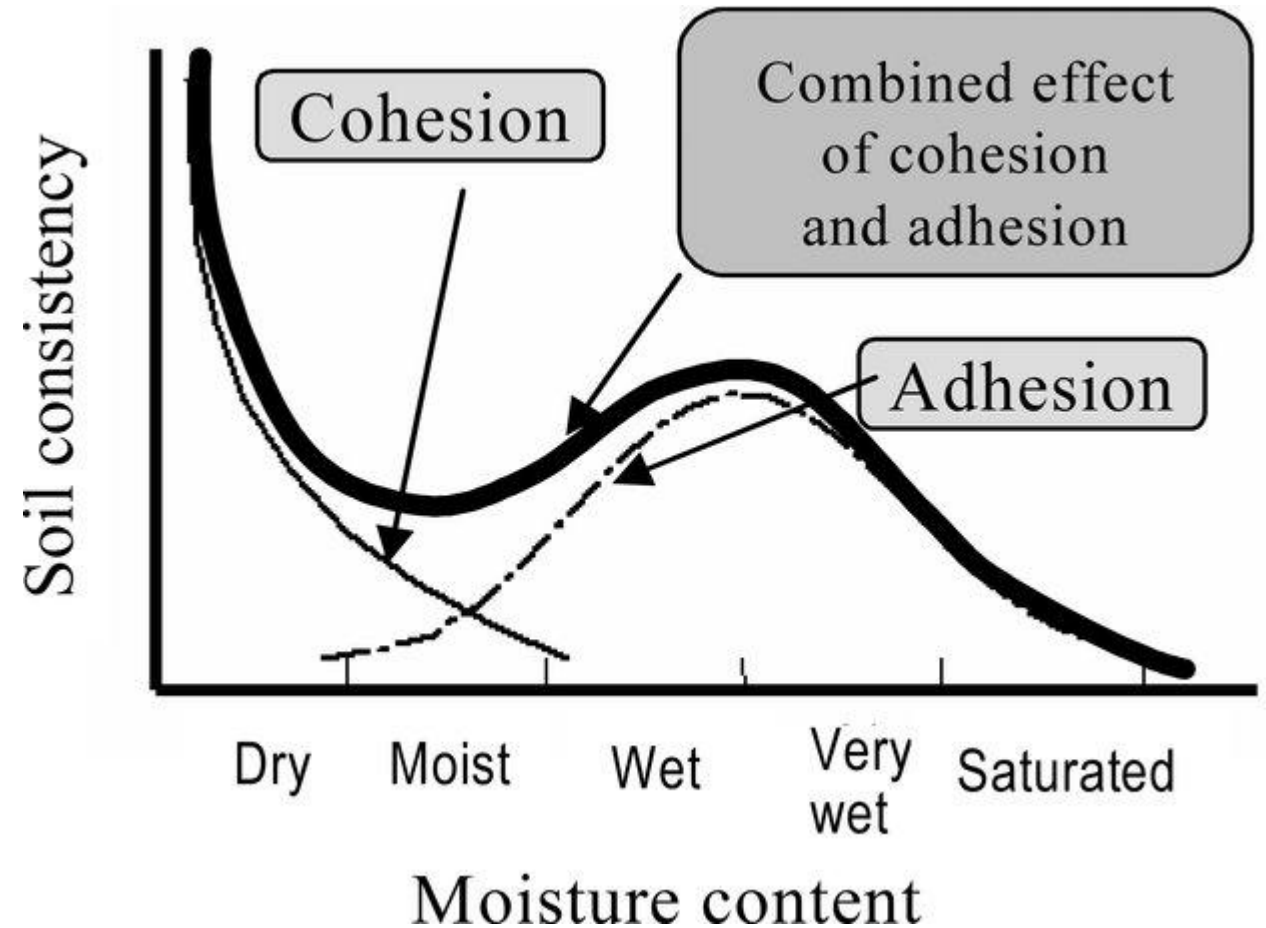


Physical properties of soils

Adhesion: molecular attraction between 2 different materials

Cohesion: attraction between molecules of the same type.

Both are electrostatic forces



Burrowing in Earthworms

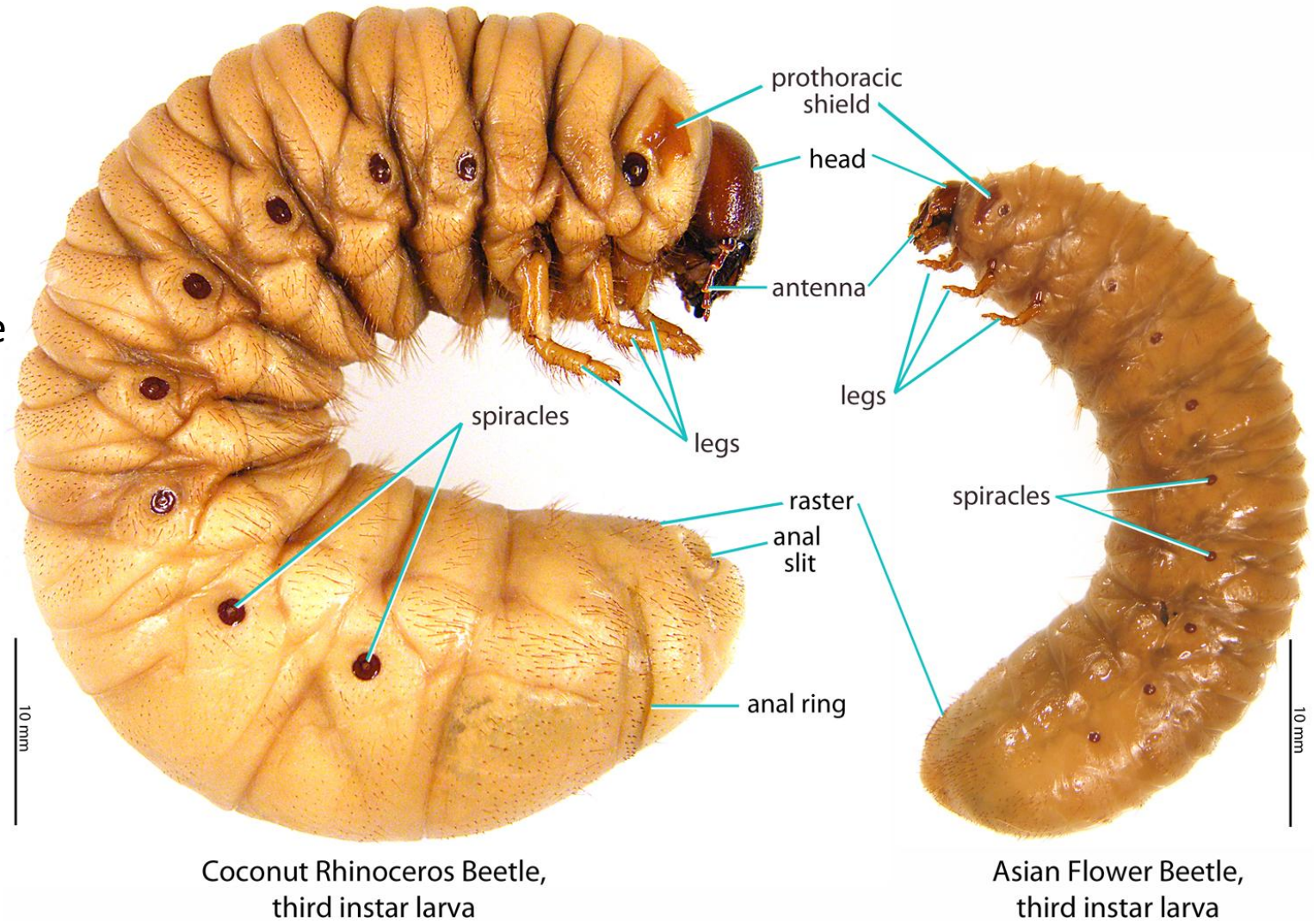
Earthworms force their front end into a slot in the soil and widen the gap by body expansion, swallowing large quantities of soil.

Mucus is also released by the gut, and is used to plaster the tunnel walls.



Burrowing in scarab beetle larvae

General morphology of Scarabaeidae larvae



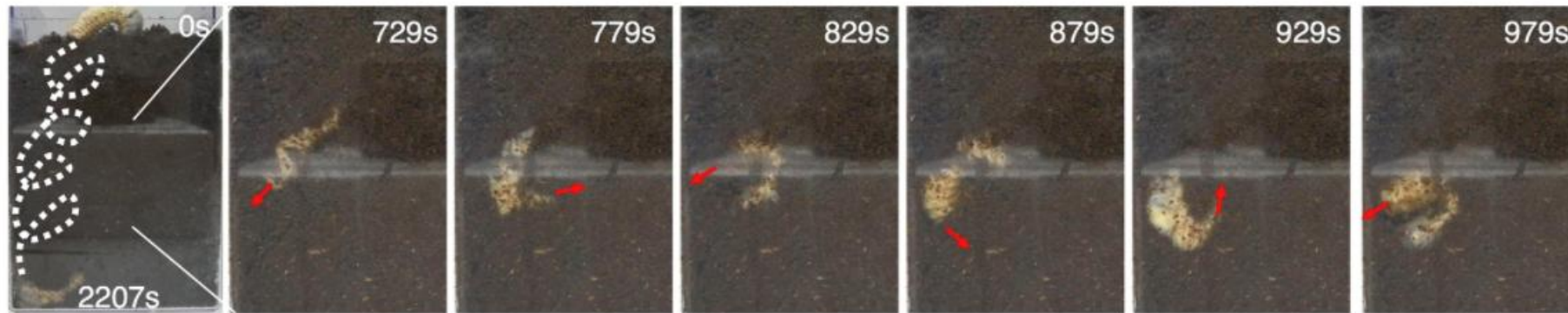
Burrowing in scarab beetle larvae

Trypoxylus dichotomus larvae burrow in two different ways, depending on the hardness of the soil.

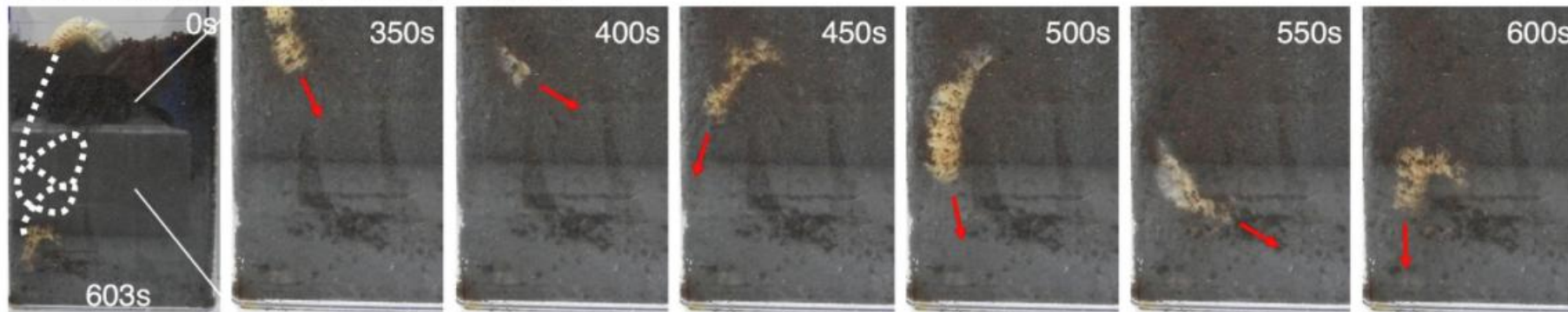
If the soil is soft, the larvae keep their body in a straight line and use longitudinal expansion and contraction.

If the soil is hard, they flex and rotate their body, adapting to diverse soil conditions using two different excavation methods. These results are important for understanding the soil ecology and pose a challenge to engineer of newer excavation technology.

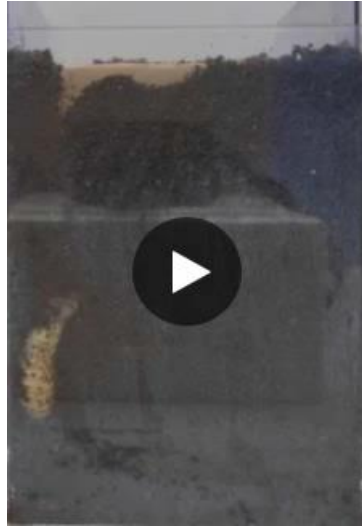
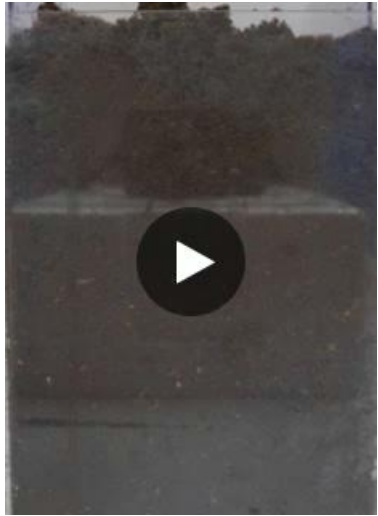
Hard condition



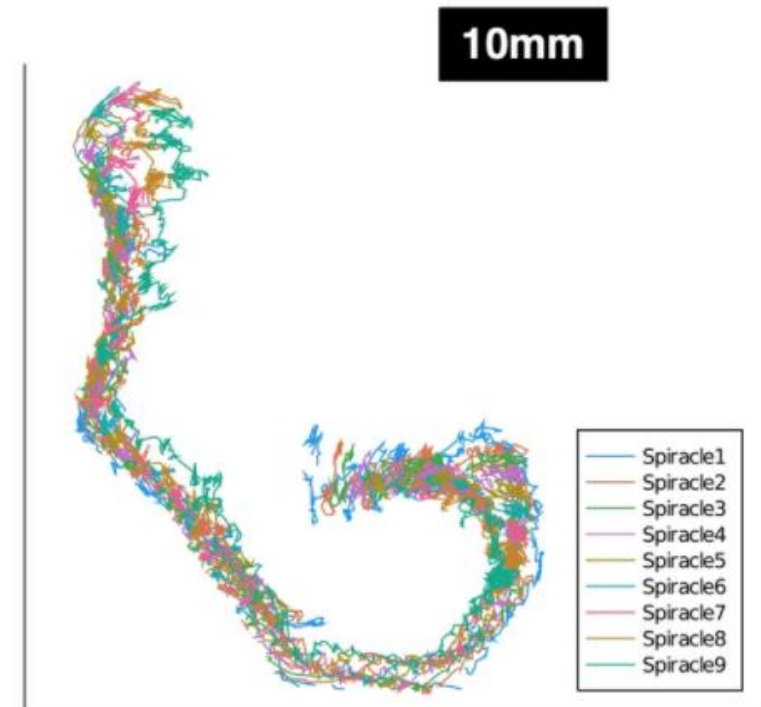
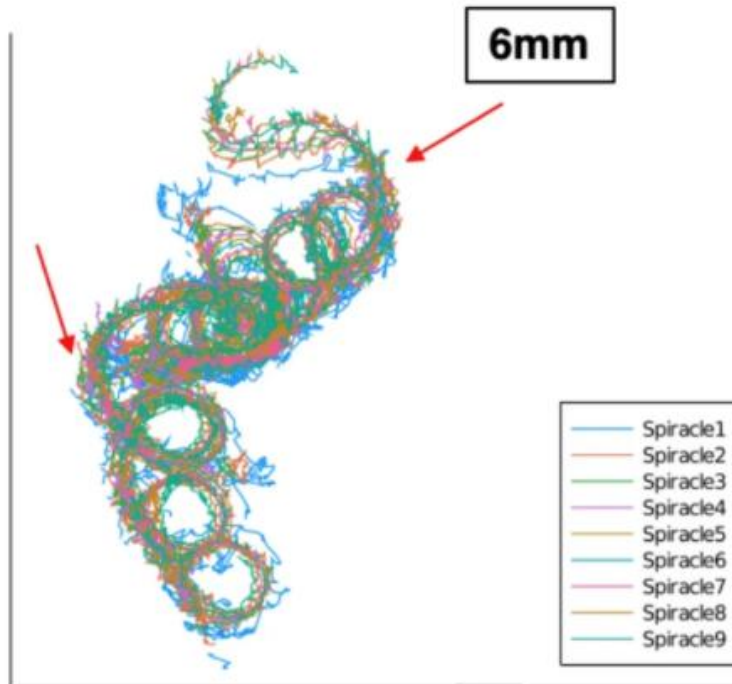
Soft condition

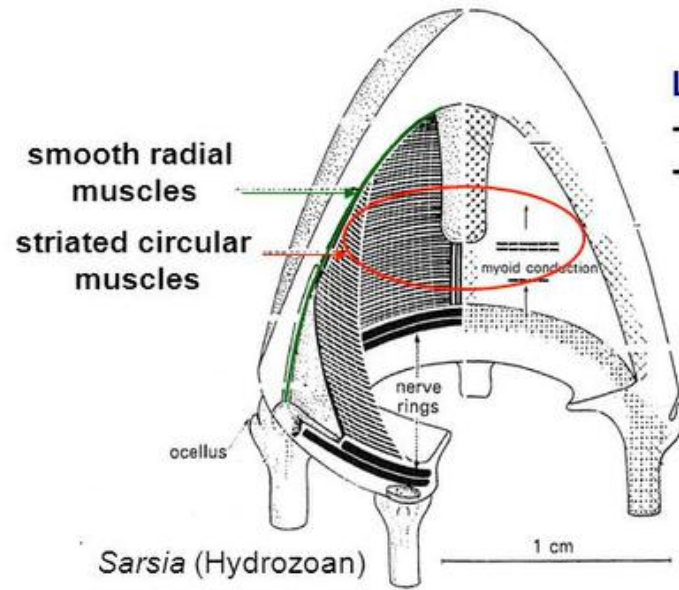


Burrowing in scarab beetle larvae



Burrowing dynamics changed when the hardness and size of the soil particles were changed. In addition, the relationship between larval shape and timing of rotation was clarified, suggesting that rotated movement may move the soil downwards.





Locomotion:

- contraction of the circular muscles
- shape recovery due to the bell elasticity